Datasets
Signals we explain

Synthetic - ground truth

- `gauss`
- `burr`
- `skeleton`
- `gait`
- `skel+gait`

Real - no ground truth

- `CARE-PD:` human motion
- `PTB-XL:` cardiac
- `ESC-50:` audio

Players $\mathcal{P}$
The n sets of features that will make up the explanation

Spatial - by spatial component



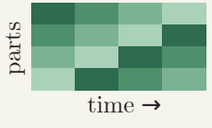
$\mathcal{P}_{\text{joint}}, M = J$
(J = joint)

Temporal - by time frame



$\mathcal{P}_{\text{temp}}, M = K$
(K = time windows)

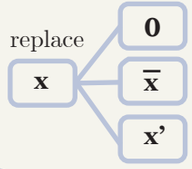
Spatiotemporal - by both



$\mathcal{P}_{\text{cell}}, M = J \times K$

Methods
How we replace a player to explain its impact (KS: Kernel SHAP)

Baselines (off-manifold)



KS-Zero, KS-Mean, KS Empirical, KS-Marginal

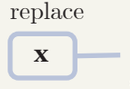
On Manifold



spatiotemporally plausible completion

KS VAEAC, KS Flow, KS Gauss

Temporal



temporally plausible completion

WindowSHAP, TimeSHAP

Metrics
How explanation quality is evaluated

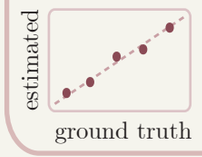
With Ground Truth

Error Criterion 1 ↓ (MACE)



distance to true credit values

Error Criterion 3 ↓

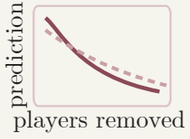


agreement of credit rankings (1 - corr. w/ true)

Without Ground Truth

Faithfulness ↑ effect of removing top players

Player Area Over the Perturbation Curve ↑



how fast that effect accrues